R21

Code No.: U21PC602CS

M V S R ENGINEERING COLLEGE (Autonomous)

B.E. VI Semester (Main & Supple.) (Branch: CSE & CSE(IOT-CS-BCT))

Examination, AUGUST 2025

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 hours Max. Marks : 70

Note : i) FIRST Question is compulsory and answer any **FOUR** questions from the remaining six questions. Each question carries 14 Marks.
ii) Answers to each question must be written at one place only and in the same order as they occur in the question paper.
iii) Missing data, if any, may suitably be assumed.

Q.No.		Marks	CO	BT levels
1. a)	Define Artificial Intelligence.	(2)	1	1
b)	List the agent types.	(2)	1	1
c)	Give me example of BBN.	(2)	2	4
d)	Explain KNN algorithm.	(2)	3	2
e)	List out the applications of Reinforcement Learning	(2)	3	1
f)	Mention the types of hierarchical clustering.	(2)	5	1
g)	Write a short note on perceptron.	(2)	4	1

2	(a)	Explain BFS with an example?	(7)	1	2
	(b)	Discuss the steps in problem formation with example.	(7)	1	2
3	(a)	Explain resolution refutation in propositional logic with example.	(7)	2	2
	(b)	Draw the Expert systems architecture. Explain in detail.	(7)	2	2
4	(a)	What are the differences between supervised and unsupervised learning.	(7)	3	2
	(b)	Build a simple linear Regression with example.	(7)	3	5
5	(a)	Sketch and explain Multi neural network with example.	(7)	4	2
	(b)	Explain classifier's performance metrics.	(7)	4	2
6	(a)	Demonstrate K-means algorithm with example.	(7)	5	2
	(b)	Explain Reinforcement learning concept with the help of example.	(7)	5	2
7	(a)	Explain Naïve Bayes classification.	(5)	3	2
	(b)	Discuss the regression evaluation metrics.	(5)	4	2
	(c)	Summarize the applications of machine learning.	(4)	5	2

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